# Retail Sales Performance Analysis

*Power BI Business Analyst Portfolio Project*

## 1. Business Problem

Management lacks a clear overview of retail sales performance across different products and cities. This makes it difficult to identify high-revenue areas and opportunities for business improvement.

## 2. Project Objective

The objective of this project is to analyse retail sales data using Power BI, monitor key performance indicators, compare sales performance across products and cities, and provide data-driven recommendations to support management decisions.

## 3. Dataset Description

This project uses a retail sales dataset containing six sales transactions across three products (Laptop, Mobile, and Headphones) and three cities (London, Manchester, and Birmingham). The dataset includes product names, city locations, and sales amounts. This information was used to analyse sales performance across products and cities.

## 4. Tools and Technologies Used

Microsoft Power BI Desktop was used to develop an interactive retail sales dashboard. DAX (Data Analysis Expressions) measures were created to calculate key performance indicators, including Total Sales, Average Sales, and Number of Sales. Charts were used to compare sales performance across products and cities, while interactive slicers allowed users to filter the results.

## 5. Key Business Findings

The retail sales analysis identified total revenue of £10,000 across six transactions. Laptops generated the highest product revenue at £5,000, representing 50% of total

sales. London was the highest-revenue city, generating £4,500, representing 45% of total sales. These findings highlight the strongest revenue contributors within the available dataset.

## 6. Business Recommendations

Based on the analysis, I recommend investigating the factors contributing to lower revenue from Headphones and in Manchester. Further analysis of customer demand, product availability, and marketing effectiveness could help management identify opportunities to improve sales performance. These recommendations should be evaluated using additional business data before making major decisions.

## 7. Project Limitations

This project was based on a small dataset containing only six sales transactions. The dataset did not include product costs, historical sales records, or customer information. As a result, the analysis could not assess profitability, identify sales trends over time, or examine customer purchasing behaviour. Additional data would be required to provide deeper business insights.

## 8. Future Project Improvements

In the future, this project could be improved by incorporating historical sales data, product costs, and customer information. Historical data would allow the analysis of monthly sales trends and revenue growth. Product cost information would enable profitability analysis, while customer data could provide insights into purchasing behaviour. These improvements would help management make more informed business decisions.

## 9. Project Conclusion

This project demonstrates how Power BI can be used to analyse retail sales data and present key business performance indicators through an interactive dashboard. The analysis identified the highest-revenue products and cities and provided evidence-based recommendations for further investigation. Although the dataset was limited, the project demonstrates practical skills in data analysis, DAX calculations, dashboard development, and communicating business insights to support management decisions.

## 10. Power BI Dashboard Screenshot.

